

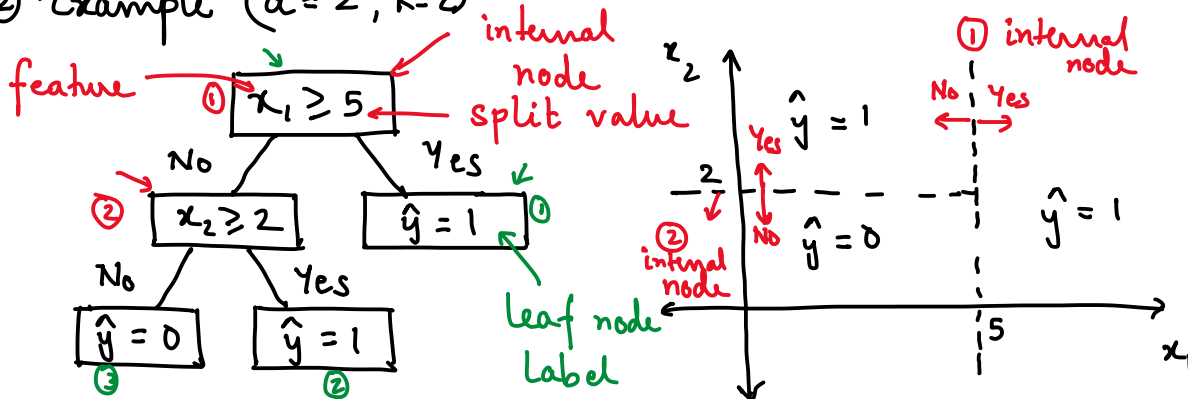
Lecture 20

Saturday, April 25, 2026 9:20 AM

Recap: Decision Trees for Classification

① Training data : $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$ $x^{(i)} \in \mathbb{R}^d$
 $y^{(i)} \in \{0, 1, 2, \dots, K-1\}$
 $x^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \end{bmatrix}$

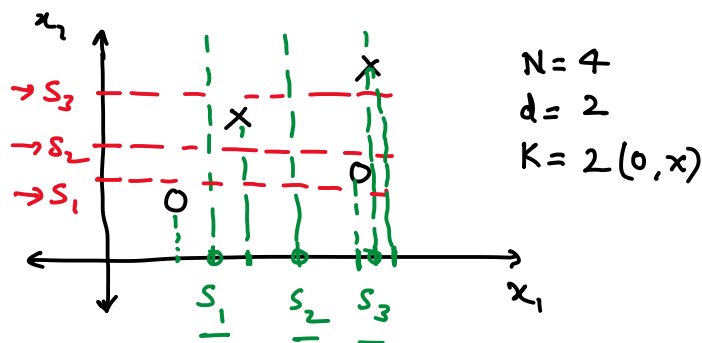
② Example ($d=2, K=2$)



③ Split values : infinite options for each feature (problem)

- Sort distinct values taken by the feature in the training data (say $v_1 < v_2 < \dots < v_m$)
- Consider only $m-1$ candidate split values,

$$s_i = \frac{v_i + v_{i+1}}{2}$$



④ Gini Impurity

Set S , $p_k = \frac{|S_k|}{|S|}$ (fraction of class k samples in S)

$$\sum_{k=0}^{K-1} p_k = 1$$

$$G(S) = \sum_{k=0}^{K-1} p_k (1 - p_k) = 1 - \sum_{k=0}^{K-1} p_k^2$$

Gini Impurity Index

$$G(S) = p_0(1-p_0) + p_1(1-p_1)$$

$$K=2 \quad p_0 = 1 - p_1 \quad (\because p_0 + p_1 = 1)$$

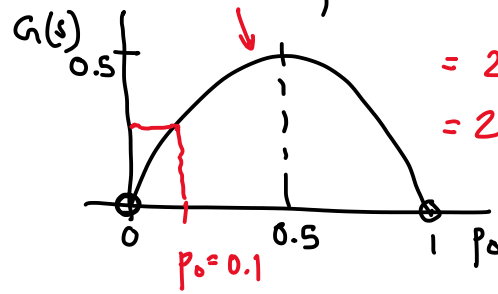
$$G(s) = \frac{2p_0(1-p_0)}{}$$

only for $K=2$

$p_0 = 0$ - all class 1

$p_0 = 1$ - all class 0

$p_0 = 1/2$ ($p_1 = 1/2$)



$$\begin{aligned} &= p_0 p_1 + p_1 p_0 \\ &= 2p_0 p_1 \\ &= 2p_0(1-p_0) \end{aligned}$$

Today:

How to build a tree from training data?

- Greedy, recursive process

Unknown

- How many internal nodes?
- What feature, split for internal node?
- Labels of leaf node?

One issue with decision trees: Overfitting

- Pre-pruning (early stopping)

Option 1: $|I| < k$ or I is pure $\leftarrow$ in our code

Option 2: is the tree depth $> D$ $\leftarrow$

Option 3: I - Gini index $\leftarrow$

I^+, I^- - weighted $G I$

If weighted $G I$ of $I^+, I^- \ll G I(I)$

- Post-pruning

a) build large first

b) Define the cost of a tree

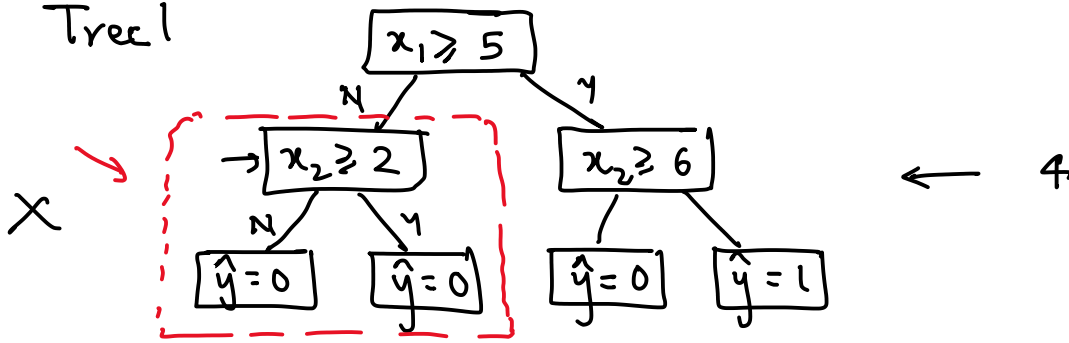
$$\rightarrow C_\alpha(T) = (\text{training error}) + \alpha (\text{number of leaves})$$

tree
parameter

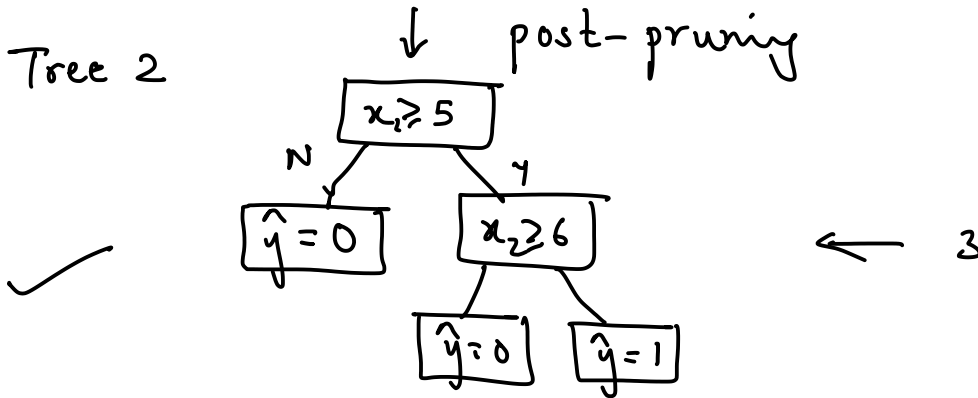
Remove leaves iteratively, to get a tree with

The lowest $C_\alpha(T)$ value.

Tree 1



Tree 2



α : cross-validation

- large tree
- define $C_\alpha(T)$ cost function
- efficient way to ~~prune~~ prune tree to get a subtree that minimize $C_\alpha(T)$

details omitted

- α chosen by crossvalidation

